State Space Methods for Flight Vehicles

AE 695

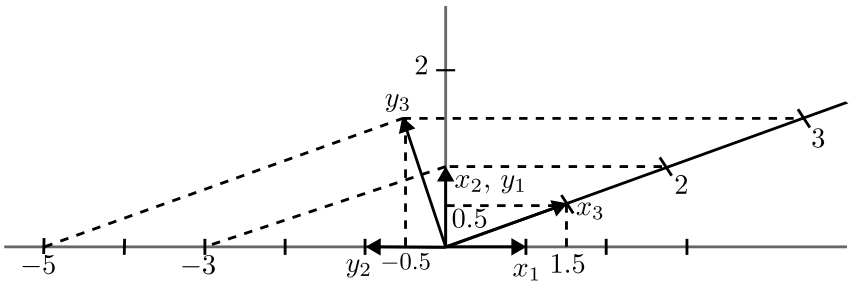
Shashi Ranjan Kumar

Associate Professor
Intelligent Systems and Control (ISaC) Lab
Department of Aerospace Engineering
Indian Institute of Technology Bombay
Powai, Mumbai- 400076

https://www.aero.iitb.ac.in/~shashi
srk@aero.iitb.ac.in

Linear Dependence: Linear Operators

- Transformation that rotates a point in the geometric plane CCW by 90° w.r.t. the origin



- With $\{\boldsymbol{x}_1, \boldsymbol{x}_2\}$ as basis

$$\boldsymbol{y}_1 = L\boldsymbol{x}_1 = \begin{bmatrix} \boldsymbol{x}_1 & \boldsymbol{x}_2 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\boldsymbol{y}_2 = L\boldsymbol{x}_2 = \begin{bmatrix} \boldsymbol{x}_1 & \boldsymbol{x}_2 \end{bmatrix} \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

- Representation of the operator L w.r.t. the basis $\{\boldsymbol{x}_1, \boldsymbol{x}_2\}$

$$\boldsymbol{A} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

- Representation of $\boldsymbol{x}_3$ w.r.t. the basis $\{\boldsymbol{x}_1, \boldsymbol{x}_2\}$

$$\boldsymbol{x}_3 = \begin{bmatrix} 1.5 \\ 0.5 \end{bmatrix}$$

- What would be its representation $\boldsymbol{y}_3$ after the rotation?



Linear Operators: Examples

- Representation of $\boldsymbol{y}_3$ w.r.t. the basis $\{\boldsymbol{x}_1, \boldsymbol{x}_2\}$

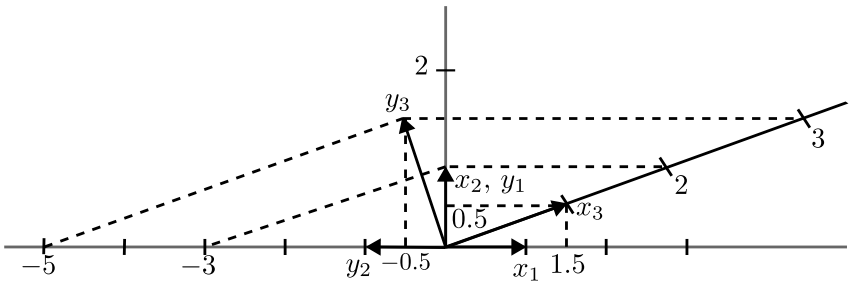
$$\boldsymbol{y}_3 = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1.5 \\ 0.5 \end{bmatrix} = \begin{bmatrix} -0.5 \\ 1.5 \end{bmatrix}$$

$$\boldsymbol{y}_3 = \begin{bmatrix} \boldsymbol{x}_1 & \boldsymbol{x}_2 \end{bmatrix} \begin{bmatrix} -0.5 \\ 1.5 \end{bmatrix}$$

$$\boldsymbol{y}_1 = L\boldsymbol{x}_1 = \begin{bmatrix} \boldsymbol{x}_1 & \boldsymbol{x}_3 \end{bmatrix} \begin{bmatrix} -3 \\ 2 \end{bmatrix}$$

- What would happen if $\{\boldsymbol{x}_1, \boldsymbol{x}_3\}$ is chosen as the basis?

$$\boldsymbol{y}_3 = L\boldsymbol{x}_3 = \begin{bmatrix} \boldsymbol{x}_1 & \boldsymbol{x}_3 \end{bmatrix} \begin{bmatrix} -5 \\ 3 \end{bmatrix}$$



- Representation of operator w.r.t. the basis $\{\boldsymbol{x}_1, \boldsymbol{x}_3\}$

$$\boldsymbol{A} = \begin{bmatrix} -3 & -5 \\ 2 & 3 \end{bmatrix}$$

Linear Operators: Examples

- Matrix $\boldsymbol{A}$: Can be considered as a linear operator that maps $(\mathcal{F}^n, \mathcal{F})$ to itself.
- For the basis $\{\boldsymbol{n}_i\}$, the representation of linear operator $\boldsymbol{A}$ is $\boldsymbol{A}$ itself.
- For the basis $\{\boldsymbol{q}_i\}$, the representation of linear operator $\boldsymbol{A}$ is

$$\bar{\boldsymbol{A}} = \boldsymbol{Q}^{-1}\boldsymbol{A}\boldsymbol{Q}, \quad \boldsymbol{Q} = \begin{bmatrix} \boldsymbol{q}_1 & \boldsymbol{q}_2 & \cdots & \boldsymbol{q}_n \end{bmatrix}$$

- Consider the matrix given by

$$L = \boldsymbol{A} = \begin{bmatrix} 3 & 2 & -1 \\ -2 & 1 & 0 \\ 4 & 3 & 1 \end{bmatrix}, \boldsymbol{b} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\boldsymbol{A}\boldsymbol{b} = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \ \boldsymbol{A}^2\boldsymbol{b} = \begin{bmatrix} -4 \\ 2 \\ -3 \end{bmatrix}, \ \boldsymbol{A}^3\boldsymbol{b} = \begin{bmatrix} -5 \\ 10 \\ -13 \end{bmatrix}$$

$$\boxed{\boldsymbol{A}^3\boldsymbol{b} = 5\boldsymbol{A}^2\boldsymbol{b} - 15\boldsymbol{A}\boldsymbol{b} + 17\boldsymbol{b}}$$

- As $\boldsymbol{b}, \boldsymbol{A}\boldsymbol{b}, \boldsymbol{A}^2\boldsymbol{b}$ are linearly independent, they can be a basis.



Linear Operators: Examples

- How to compute the representation of the operator $\boldsymbol{A}$ w.r.t this basis?

$$\boldsymbol{A}(\boldsymbol{b}) = \begin{bmatrix} \boldsymbol{b} & \boldsymbol{Ab} & \boldsymbol{A^2 b} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\boldsymbol{A}(\boldsymbol{Ab}) = \begin{bmatrix} \boldsymbol{b} & \boldsymbol{Ab} & \boldsymbol{A^2 b} \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\boldsymbol{A}(\boldsymbol{A^2 b}) = \begin{bmatrix} \boldsymbol{b} & \boldsymbol{Ab} & \boldsymbol{A^2 b} \end{bmatrix} \begin{bmatrix} 17 \\ -15 \\ 5 \end{bmatrix}$$

- Representation of linear operator $\boldsymbol{A}$ w.r.t. $\{\boldsymbol{b}, \boldsymbol{Ab}, \boldsymbol{A^2 b}\}$ as the basis

$$\bar{\boldsymbol{A}} = \begin{bmatrix} 0 & 0 & 17 \\ 1 & 0 & -15 \\ 0 & 1 & 5 \end{bmatrix} = \boldsymbol{Q}^{-1}\boldsymbol{A}\boldsymbol{Q}$$



Linear Operators: Examples

- Assume $n \times n$ matrix with real coefficients, and $\exists\, \boldsymbol{b}$ s.t. $\{\boldsymbol{b}, \boldsymbol{A}\boldsymbol{b}, \ldots, \boldsymbol{A}^{n-1}\boldsymbol{b}\}$ are linearly independent and

$$\boldsymbol{A}^n\boldsymbol{b} = -\alpha_n\boldsymbol{b} - \alpha_{n-1}\boldsymbol{A}\boldsymbol{b} - \cdots - \alpha_1\boldsymbol{A}^{n-1}\boldsymbol{b}$$

- Representation of matrix $\boldsymbol{A}$ w.r.t. the basis $\boldsymbol{b}, \boldsymbol{A}\boldsymbol{b}, \ldots, \boldsymbol{A}^{n-1}\boldsymbol{b}$

$$\bar{\boldsymbol{A}} = \begin{bmatrix} 0 & 0 & \cdots & 0 & -\alpha_n \\ 1 & 0 & \cdots & 0 & -\alpha_{n-1} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & -\alpha_2 \\ 0 & 0 & \cdots & 1 & -\alpha_1 \end{bmatrix}$$

- A companion form of linear operator
- Using $\bar{\boldsymbol{A}} = \boldsymbol{Q}^{-1}\boldsymbol{A}\boldsymbol{Q}$, we have

$$\boldsymbol{Q}\bar{\boldsymbol{A}} = \boldsymbol{A}\boldsymbol{Q} \implies \begin{bmatrix}\boldsymbol{q}_1 & \boldsymbol{q}_2 \cdots & \boldsymbol{q}_n\end{bmatrix}\bar{\boldsymbol{A}} = \begin{bmatrix}\boldsymbol{A}\boldsymbol{q}_1 & \boldsymbol{A}\boldsymbol{q}_2 \cdots & \boldsymbol{A}\boldsymbol{q}_n\end{bmatrix}$$

- i^{th} column of $\bar{\boldsymbol{A}}$: Representation of $\boldsymbol{A}\boldsymbol{q}_i$ w.r.t. basis $\boldsymbol{q}_1, \boldsymbol{q}_2, \boldsymbol{q}_3, \ldots, \boldsymbol{q}_n$



Systems of Linear Algebraic Equations

- System of linear algebraic equations

$$
\begin{aligned}
a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= y_1 \\
a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= y_2 \\
\cdots\cdots\cdots

Systems of Linear Algebraic Equations

- Important questions:
  - Existence of a solution and number of solutions
- What would be the condition on $\boldsymbol{A}, \boldsymbol{y}$ so that $\exists\ \boldsymbol{x}$ s.t. $\boldsymbol{Ax} = \boldsymbol{y}$?
- How many l.i. vectors $\boldsymbol{x}$ exists such that $\boldsymbol{Ax} = \boldsymbol{y}$?

Range of linear operator $\boldsymbol{A}$

$$\mathcal{R}(\boldsymbol{A}) = \{\boldsymbol{y} \in (\mathcal{F}^m, \mathcal{F}) |\ \exists\ \boldsymbol{x} \in (\mathcal{F}^n, \mathcal{F}),\quad \boldsymbol{y} = \boldsymbol{Ax}\}$$

The range of a linear operator is a subspace of $(\mathcal{F}^m, \mathcal{F})$.

- If $\boldsymbol{y}_1, \boldsymbol{y}_2 \in \mathcal{R}(\boldsymbol{A})\ \exists\ \boldsymbol{x}_1, \boldsymbol{x}_2 \in (\mathcal{F}^n, \mathcal{F})$ s.t. $\boldsymbol{y}_1 = \boldsymbol{Ax}_1,\quad \boldsymbol{y}_2 = \boldsymbol{Ax}_2$
- **Claim**: For any α_1, α_2, $\alpha_1\boldsymbol{y}_1 + \alpha_2\boldsymbol{y}_2 \in \mathcal{R}(\boldsymbol{A})$,

$$\alpha_1\boldsymbol{y}_1 + \alpha_2\boldsymbol{y}_2 = \boldsymbol{A}(\alpha_1\boldsymbol{x}_1 + \alpha_2\boldsymbol{x}_2) \implies \alpha_1\boldsymbol{x}_1 + \alpha_2\boldsymbol{x}_2 \in (\mathcal{F}^n, \mathcal{F})$$

- Thus, the range $\mathcal{R}(\boldsymbol{A})$ is a subspace of $(\mathcal{F}^m, \mathcal{F})$.



- If $\boldsymbol{A} = \begin{bmatrix} \boldsymbol{a}_1 & \boldsymbol{a}_2 & \cdots & \boldsymbol{a}_n \end{bmatrix}$

$$\boldsymbol{y} = x_1\boldsymbol{a}_1 + x_2\boldsymbol{a}_2 + \cdots + x_n\boldsymbol{a}_n, \; x_i \in \mathcal{F}$$

- $\mathcal{R}(\boldsymbol{A})$: Set of all the possible linear combinations of the columns of $\boldsymbol{A}$.
- Dimension of $\mathcal{R}(\boldsymbol{A})$
  - Maximum number of l.i. vectors in $\mathcal{R}(\boldsymbol{A})$
  - Maximum number of l.i. columns in $\boldsymbol{A}$

Rank of a Matrix $\rho(\boldsymbol{A})$

Maximum number of l.i. columns in matrix $\boldsymbol{A}$, the dimension of the range space $\mathcal{R}(\boldsymbol{A})$.

- How to compute the rank of a matrix?
- Using a sequence of elementary transformations
- No change in the rank with pre- or post-multiplications by elementary matrices



Elementary Transformation

- What do you mean by elementary transformation or operations?
  - Multiplying a row or a column by a nonzero real or complex number
  - Interchanging two rows or two columns
  - Adding the product of one row or column and a number to another row or column
- How to achieve these operations through elementary matrices?
- Using elementary matrices, with $c \neq 0 \in \mathbb{R}, \mathbb{C}$

$$\boldsymbol{E}_1 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & c & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \boldsymbol{E}_2 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 \end{bmatrix}, \boldsymbol{E}_3 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & d & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\boldsymbol{E}_1^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1/c & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \boldsymbol{E}_2^{-1} = \boldsymbol{E}_2, \boldsymbol{E}_3^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & -d & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$



Gaussian Elimination

- How to perform the Gaussian elimination on the matrix $\boldsymbol{A}$?

$$\boldsymbol{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1m} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2m} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nm} \end{bmatrix}$$

- Using elementary matrices, with $e_{i1} = -\dfrac{a_{i1}}{a_{11}}$, with $a_{11} \neq 0$

$$\boldsymbol{K}_1\boldsymbol{A} = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ e_{21} & 1 & 0 & \cdots & 0 \\ e_{31} & 0 & 1 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ e_{n1} & 0 & 0 & 0 & 1 \end{bmatrix} \boldsymbol{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1m} \\ 0 & a^1_{22} & a^1_{23} & \cdots & a^1_{2m} \\ 0 & a^1_{32} & a^1_{33} & \cdots & a^1_{3m} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & a^1_{n2} & a^1_{n3} & \cdots & a^1_{nm} \end{bmatrix}$$

- $\boldsymbol{K}_1$: Product of $n-1$ elementary matrices of the form E_3
- Entries of resultant matrix

$$\boxed{a^1_{ij} = a_{ij} + e_{i1}a_{1j}}$$



- For $a^1_{22} \neq 0$, with $e_{i2} = -\dfrac{a^1_{i2}}{a^1_{22}}$, with $a^1_{22} \neq 0$,

$$\boldsymbol{K_2 K_1 A} = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & e_{32} & 1 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & e_{n2} & 0 & 0 & 1 \end{bmatrix} \boldsymbol{K_1 A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1m} \\ 0 & a^1_{22} & a^1_{23} & \cdots & a^1_{2m} \\ 0 & 0 & a^2_{33} & \cdots & a^2_{3m} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & a^2_{n3} & \cdots & a^2_{nm} \end{bmatrix}$$

- $\boldsymbol{K_2}$: Product of elementary matrices

$$\boxed{a^2_{ij} = a^1_{ij} + e^1_{i2} a^1_{2j}}$$

- Upper right triangular matrices using Gaussian elimination
- Under what condition will Gaussian elimination be unstable?
- How to avoid such issues?
- Gaussian elimination with partial pivoting or complete pivoting methods
- How to perform Gaussian elimination with these methods?



Gaussian Elimination using Partial or Complete Pivoting

- **Gaussian elimination with partial pivoting**:
  - Search the element with the largest absolute value in the first column
  - Interchange this row with the first row using elementary transformation
  - Carry out the elimination of the first column
  - Search for the largest element in the second column of $\boldsymbol{K}_1\boldsymbol{A}$, excluding first element and bring to a_{22}^1 position
  - Process is more stable as $|e_{ij}| \leq 1$
- **Gaussian elimination with complete pivoting**:
  - Use of both elementary row and column operations
  - Search the element with the largest absolute value among all elements
  - Move the element to the first column by the elementary column operation and then to the first row by elementary row operation
  - Carry out the elimination of the first column
  - Repeat the process with the remaining matrix
  - Process is more stable than partial pivoting, but at the cost of time-consuming processing
- Partial pivoting more popular w.r.t. overall performance (efficiency, accuracy, reliability, generality, ease of use, etc)



Rank of Matrix

- **Upper triangular matrix**: Rank, same as number of nonzero rows
- For a matrix $\boldsymbol{A}_{m\times n}$,

$$\text{rank}(\boldsymbol{A}) = \text{No. of l.i. columns} = \text{No. of l.i. rows} \leq (\min(n, m))$$

- **Rank of a matrix**: Largest order of non-vanishing minors of a matrix
- A matrix is of rank k iff there is at least one minor of order k that does not vanish and every minor of order higher than k vanishes.
- A square matrix is of full rank iff its determinant is nonzero.
- A square matrix is nonsingular iff all the rows and columns are linearly independent.
- Examples:

$$\boldsymbol{A} = \begin{bmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 0 & 2 & 0 \end{bmatrix}, \quad \boldsymbol{B} = \begin{bmatrix} 1 & 2 \\ 0 & \epsilon \end{bmatrix}$$

- Second matrix is ambiguous in some sense, **need to check the source of ϵ**



Solution of Linear Equations

Consider the matrix equation $\boldsymbol{Ax} = \boldsymbol{y}$, $\boldsymbol{A} : (\mathcal{F}^n, \mathcal{F}) \rightarrow (\mathcal{F}^m, \mathcal{F})$.

- Given $\boldsymbol{A}$ and $\boldsymbol{y}$ in $(\mathcal{F}^m, \mathcal{F})$, there exists a vector $\boldsymbol{x}$ such that $\boldsymbol{Ax} = \boldsymbol{y}$ **iff** the vector $\boldsymbol{y}$ is an element of $\mathcal{R}(\boldsymbol{A})$, $\rho(\boldsymbol{A}) = \rho([\boldsymbol{A} : \boldsymbol{y}])$.
- Given $\boldsymbol{A}$, for every $\boldsymbol{y}$ in $(\mathcal{F}^m, \mathcal{F})$, there exists a vector $\boldsymbol{x}$ such that $\boldsymbol{Ax} = \boldsymbol{y}$ **iff** $\mathcal{R}(\boldsymbol{A}) = (\mathcal{F}^m, \mathcal{F})$ or equivalently, $\rho(\boldsymbol{A}) = m$.

- Follows from the definition of range space.
- If $\boldsymbol{y} \notin \mathcal{R}(\boldsymbol{A})$, $\boldsymbol{Ax} = \boldsymbol{y}$ has no solution and is said to be inconsistent.
- $\rho(\boldsymbol{A})$: Dimension of $\mathcal{R}(\boldsymbol{A})$
- Since $\mathcal{R}(\boldsymbol{A})$ is subspace of $(\mathcal{F}^m, \mathcal{F})$, $\rho(\boldsymbol{A}) = m \implies \mathcal{R}(\boldsymbol{A}) = (\mathcal{F}^m, \mathcal{F})$.
- If $\mathcal{R}(\boldsymbol{A}) = (\mathcal{F}^m, \mathcal{F})$, then for every $\boldsymbol{y} \in (\mathcal{F}^m, \mathcal{F})$, $\exists\, \boldsymbol{x}$ s.t. $\boldsymbol{Ax} = \boldsymbol{y}$.
- If $\rho(\boldsymbol{A}) < m$, $\exists$ at least one nonzero vector $\boldsymbol{y} \in (\mathcal{F}^m, \mathcal{F})$, but $\boldsymbol{y} \notin \mathcal{R}(\boldsymbol{A})$ for which there exist no $\boldsymbol{x}$ s.t. $\boldsymbol{Ax} = \boldsymbol{y}$.



Full Row/Column Rank and Null Space

- **Full row rank**: If all rows of the matrix $m \times n$ are l.i.
- **Full column rank**: If all columns of the $m \times n$ matrix are l.i.

Null Space of linear operator $\mathcal{N}(\boldsymbol{A})$

All the elements $\boldsymbol{x} \in (\mathcal{F}^n, \mathcal{F})$ for which $\boldsymbol{Ax} = \boldsymbol{0}$.
The dimension of null space is the nullity of $\boldsymbol{A}$, denoted by $\nu(\boldsymbol{A})$.

- **Left and right null spaces**: $\boldsymbol{yA} = \boldsymbol{0}$ or $\boldsymbol{Ax} = \boldsymbol{0}$
- $\mathcal{N}(\boldsymbol{A})$ is a linear space.
- If $\nu(\boldsymbol{A}) = 0$, $\mathcal{N}(\boldsymbol{A})$ has only a zero vector and $\boldsymbol{Ax} = \boldsymbol{0}$ has only zero vector as solution.
- If $\nu(\boldsymbol{A}) = k$, $\mathcal{N}(\boldsymbol{A})$ then $\boldsymbol{Ax} = \boldsymbol{0}$ has k l.i. vector solutions.
- Null space and range space are the subspaces of $(\mathcal{F}^n, \mathcal{F})$ and $(\mathcal{F}^m, \mathcal{F})$, respectively.



Example

- Given matrix

$$\boldsymbol{A} = \begin{bmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 0 & 2 & 0 \end{bmatrix}$$

- Last 2 columns l.d. on first two columns, $\rho(\boldsymbol{A}) = 2$
- With $\boldsymbol{x} = [x_1 \ x_2 \ x_3 \ x_4]^\top$, $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{0}$

$$x_2 + x_3 + 2x_4 = 0, x_1 + 2x_2 + 3x_3 + 4x_4 = 0, 2x_1 + 2x_3 = 0$$

$$\boldsymbol{x} = \begin{bmatrix} 1 \\ 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ 0 \\ -1 \end{bmatrix}$$

- Two l.i. vectors in null space, thus the nullity of $\boldsymbol{A}$= 2
- Rank + Nullity = Number of columns
- What about the matrix given below?

$$\boldsymbol{A} = \begin{bmatrix} 0 & 1 & 1 & 2 & -1 \\ 1 & 2 & 3 & 4 & -1 \\ 2 & 0 & 2 & 0 & 2 \end{bmatrix}$$



Example

- Last 3 columns l.d. on first two columns, $\rho(\boldsymbol{A}) = 2$
- With $\boldsymbol{x} = [x_1 \; x_2 \; x_3 \; x_4 \; x_5]'$,

$$\boldsymbol{A}\boldsymbol{x} = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} x_1 + \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} x_2 + \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} x_3 + \begin{bmatrix} 2 \\ 4 \\ 0 \end{bmatrix} x_4 + \begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix} x_5$$

$$\boldsymbol{A}\boldsymbol{x} = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} (x_1 + x_3 + x_5) + \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} (x_2 + x_3 + 2x_4 - x_5)$$

- Due to linear independence nature of vectors, $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{0}$ implies

$$x_1 + x_3 + x_5 = 0, x_2 + x_3 + 2x_4 - x_5 = 0$$

- Number of equations: 2; Unknown variables in solutions: 5
- Arbitrary assigned components: 3

$$\boldsymbol{x} = \begin{bmatrix} -1 \\ -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -2 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$



Rank-Nullity Theorem

Rank-Nullity Theorem

For a matrix having the dimension of $m \times n$,

$$\boxed{\rho(\boldsymbol{A}) + \nu(\boldsymbol{A}) = n}$$

- Number of l.i. vector solutions of $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{0}$ is equal to $n - \rho(\boldsymbol{A})$, where n denotes number of columns of $\boldsymbol{A}$.
- Trivial solution only if $\boxed{\rho = n}$.

$$\boldsymbol{A} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \boldsymbol{B} = \begin{bmatrix} 1 & 1 & 2 & 3 \\ 3 & 4 & -1 & 2 \\ -1 & -2 & 5 & 4 \end{bmatrix}, \boldsymbol{C} = \begin{bmatrix} 1 & 3 & 4 & 3 \\ 3 & 9 & 12 & 9 \\ 1 & 3 & 4 & 1 \end{bmatrix}$$

- Using elementary transformations, matrices $\boldsymbol{B}$ and $\boldsymbol{C}$ can be reduced to

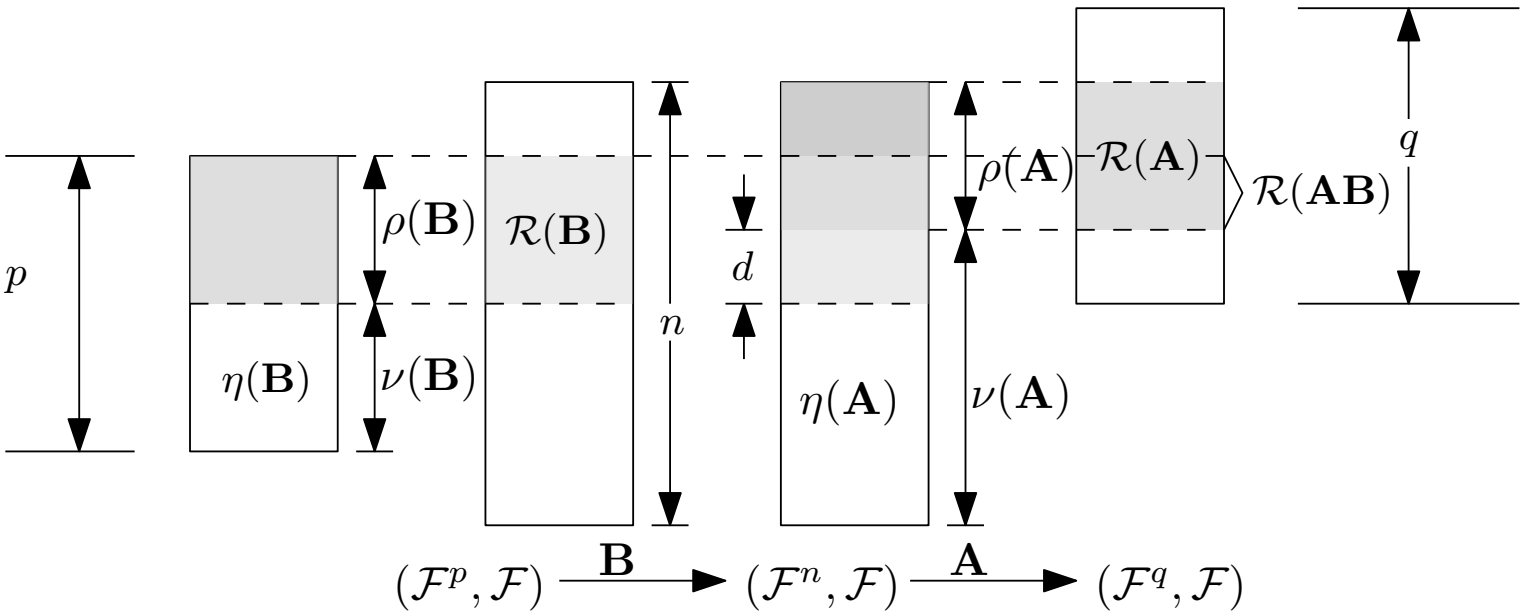
$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$



Sylvester's Inequality

Let matrices $\boldsymbol{A}$ and $\boldsymbol{B}$ be $q \times n$ and $n \times p$ with coefficients in same field.

$$\rho(\boldsymbol{A}) + \rho(\boldsymbol{B}) - n \leq \rho(\boldsymbol{AB}) \leq \min(\rho(\boldsymbol{A}), \rho(\boldsymbol{B}))$$



- **Composite matrix**: Successive application of two linear transformations
- Domain of $\boldsymbol{AB}$: $\mathcal{R}(\boldsymbol{B})$, Range of $\boldsymbol{AB}$: Subspace of $\mathcal{R}(\boldsymbol{A})$

$$\rho(\boldsymbol{AB}) \leq \min(\rho(\boldsymbol{A}), \rho(\boldsymbol{B}))$$

Sylvester's Inequality: Additional Insights

- With d as the dimension of intersection of $\mathcal{R}(\boldsymbol{B})$ and $\mathcal{N}(\boldsymbol{A})$

$$\boxed{\rho(\boldsymbol{AB}) = \rho(\boldsymbol{B}) - d}$$

$$\text{dim } \mathcal{N}(\boldsymbol{A}) = n - \rho(\boldsymbol{A}) \implies d \leq n - \rho(\boldsymbol{A})$$

- Combining the above results

$$d = \rho(\boldsymbol{B}) - \rho(\boldsymbol{AB}) \leq n - \rho(\boldsymbol{A}) \implies \boxed{\rho(\boldsymbol{AB}) \geq \rho(\boldsymbol{B}) - n + \rho(\boldsymbol{A})}$$

If the matrix $\boldsymbol{B}$ is an $n \times n$ matrix and nonsingular,

$$\rho(\boldsymbol{A}) + \rho(\boldsymbol{B}) - n = \rho(\boldsymbol{A}) \leq \rho(\boldsymbol{AB}) \leq \min(\rho(\boldsymbol{A}), n) = \rho(\boldsymbol{A})$$

For a matrix $\boldsymbol{A}$ with dimension $m \times n$, $\boxed{\rho(\boldsymbol{AC}) = \rho(\boldsymbol{A}),\ \rho(\boldsymbol{DA}) = \rho(\boldsymbol{A})}$ for any $n \times n$ and $m \times m$ nonsingular matrices $\boldsymbol{C}$ and $\boldsymbol{D}$.

- Rank of a matrix does not change with pre- or post-multiplication by a nonsingular matrix.



Matrix Properties

Let $\boldsymbol{A}$ be the $m \times n$ matrix with the coefficients in a field and $\boldsymbol{A}^\star$ be its complex conjugate transpose.

- $\rho(\boldsymbol{A}) = n$ iff $\rho(\boldsymbol{A}^\star \boldsymbol{A}) = n$, equivalently, $\det(\boldsymbol{A}^\star \boldsymbol{A}) \neq 0$.
- $\rho(\boldsymbol{A}) = m$ iff $\rho(\boldsymbol{A}\boldsymbol{A}^\star) = m$, equivalently, $\det(\boldsymbol{A}\boldsymbol{A}^\star) \neq 0$

- **Sufficiency**: To show $\boxed{\rho(\boldsymbol{A}^\star \boldsymbol{A}) = n \implies \rho(\boldsymbol{A}) = n}$

$$\boldsymbol{A\alpha} = \boldsymbol{0} \implies \boldsymbol{A}^\star \boldsymbol{A\alpha} = \boldsymbol{0} \implies \boldsymbol{\alpha} = \boldsymbol{0} \implies \rho(\boldsymbol{A}) = n$$

- **Necessity**: To show $\boxed{\rho(\boldsymbol{A}) = n \implies \rho(\boldsymbol{A}^\star \boldsymbol{A}) = n}$

$$\boldsymbol{A}^\star \boldsymbol{A\alpha} = \boldsymbol{0} \implies \boldsymbol{\alpha}^\star \boldsymbol{A}^\star \boldsymbol{A\alpha} = \boldsymbol{0}$$

- With $\boldsymbol{A\alpha} = [\beta_1, \beta_2, \cdots, \beta_n]^\top$, $\boldsymbol{\alpha}^\star \boldsymbol{A}^\star \boldsymbol{A\alpha} = \sum_{i=1}^{n} |\beta_i|^2$

$$\boldsymbol{\alpha}^\star \boldsymbol{A}^\star \boldsymbol{A\alpha} = 0 \implies \beta_i = 0 \implies \boldsymbol{A\alpha} = \boldsymbol{0}$$

- As $\rho(\boldsymbol{A}) = n$, $\boldsymbol{A\alpha} = \boldsymbol{0} \implies \boldsymbol{\alpha} = \boldsymbol{0}$, thus $\rho(\boldsymbol{A}^\star \boldsymbol{A}) = n$



Eigenvalues and Eigenvectors

Let $\boldsymbol{A}$ be a linear operator that maps $(\mathbb{C}^n, \mathbb{C})$ into itself. A scalar $\lambda \in \mathbb{C}$ is called an eigenvalue of $\boldsymbol{A}$ if there exists a nonzero vector $\boldsymbol{x}$ s.t, $\boldsymbol{A}\boldsymbol{x} = \lambda\boldsymbol{x}$. Any nonzero $\boldsymbol{x}$ satisfying $\boldsymbol{A}\boldsymbol{x} = \lambda\boldsymbol{x}$ is called an eigenvector of $\boldsymbol{A}$, associated with the eigenvalue λ.

$$\boxed{\boldsymbol{A}\boldsymbol{x} = \lambda\boldsymbol{x} \implies (\boldsymbol{A} - \lambda\boldsymbol{I})\,\boldsymbol{x} = \boldsymbol{0}}$$

- For a nontrivial solution, $\det(\boldsymbol{A} - \lambda\boldsymbol{I}) = 0$.
- λ is an eigenvalue iff it is a solution of the characteristic polynomial

$$\Delta(\lambda) \triangleq \det(\boldsymbol{A} - \lambda\boldsymbol{I}) = 0$$

- Number of eigenvalues same as the degree, n, of $\Delta(\lambda)$
- Eigenvalues are not necessarily distinct.
- **Right and left eigenvectors**: $\boldsymbol{A}\boldsymbol{x} = \lambda\boldsymbol{x}, \quad \boldsymbol{y}\boldsymbol{A} = \lambda\boldsymbol{y}$
- **Consideration of complex field**



Eigenvalues and Eigenvectors

- Consider the matrix mapping $(\mathbb{R}^2, \mathbb{R})$ to itself.

$$\boldsymbol{A} = \begin{bmatrix} 1 & -1 \\ 2 & -1 \end{bmatrix}$$

- Can we have the same results for eigenvalue and eigenvector definition if $\lambda \in \mathbb{R}$?
- For nontrivial solution of $\boldsymbol{A}\boldsymbol{x} = \lambda\boldsymbol{x}$, $\det(\lambda\boldsymbol{I} - \boldsymbol{A})$=0

$$|\lambda\boldsymbol{I} - \boldsymbol{A}| = \left| \begin{bmatrix} \lambda - 1 & 1 \\ -2 & \lambda + 1 \end{bmatrix} \right| = \lambda^2 + 1 = 0$$

- No real-values solution, thus no real eigenvalue.
- With the assumption of linear operator over $(\mathbb{C}^2, \mathbb{C})$

$$\lambda = \pm i$$

- Eigenvector with $\lambda_1 = \pm i$

$$(\boldsymbol{A} - \lambda_1\boldsymbol{I})\boldsymbol{v}_1 = \begin{bmatrix} 1 \mp i & -1 \\ 2 & -1 \mp i \end{bmatrix} \begin{bmatrix} v_{11} \\ v_{21} \end{bmatrix} = \boldsymbol{0} \implies \boxed{\boldsymbol{v}_1 = \begin{bmatrix} 1 \\ 1 \mp i \end{bmatrix}}$$



Reference

Reference

- **Chen, Chi-Tsong, "Linear System Theory and Design", Oxford University Press, 3rd Edition, 1999.**

